Ashritha Gugire

Data Scientist

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Summary

Data Scientist with 4+ years of experience in developing and deploying machine learning and deep learning models across healthcare and finance domains. Proficient in Python, SQL, PySpark, and cloud platforms like AWS, Azure, and GCP, with hands-on expertise in MLOps tools including MLflow, FastAPI, Docker, and Kubernetes. Specialized in NLP, LLMs (BERT, GPT, LangChain), time series forecasting (ARIMA, Prophet, LSTM), anomaly detection, and generative AI. Strong background in data engineering, statistical modeling, and real-time analytics using Airflow, dbt, and Databricks. Adept at translating business and regulatory requirements into scalable AI/ML solutions using Agile and CI/CD methodologies.

Experience

Deloitte, USA | Data Scientist — **April 2024 - Current**

- Developed an advanced anomaly detection system leveraging graph-based techniques, Isolation Forest, Autoencoders, and clustering methods (K-Means, DBSCAN) on datasets exceeding 60,000 records, enabling real-time issue detection in operational systems and reducing false positives by over 20%.
- Engineered and deployed ensemble models using XGBoost, Random Forest, AdaBoost, and Logistic Regression to predict user behavior with an 18% increase in accuracy, supported by rigorous statistical validation including Monte Carlo simulations and hypothesis testing.
- Built scalable MLOps pipelines with MLflow, FastAPI, Docker, and AWS EC2 for continuous deployment and monitoring of predictive models, integrating anomaly streaming workflows and performance tracking with Weights & Biases (W&B) in CI/CD environments.
- Utilized NetworkX and 15 plus knowledge graph techniques to model agent-based behavior and temporal patterns in urban or healthcare operations, applying time series methods including Prophet and LSTM for forecasting and prescriptive planning.
- Applied NLP and LLM-based frameworks using spaCy, BERT, GPT, LangChain, and Retrieval-Augmented Generation (RAG) pipelines to perform Named Entity Recognition (NER), provider note summarization, and contextual text analysis from EHR notes and clinical documents.
- Created analytical dashboards in Tableau and Power BI to visualize anomaly trends, operational metrics, and chronic disease risks, collaborating with clinicians and compliance teams to translate healthcare requirements into actionable, data-driven strategies.

Rlogical Techsoft Pvt. Ltd, India | Data Scientist — **Jan 2020 - Dec 2022**

- Developed credit scoring and risk assessment models using XGBoost, Logistic Regression, and Random Forest, improving loan approval accuracy and reducing default rates in consumer lending portfolios.
- Built real-time fraud detection systems using Isolation Forest, Autoencoders, and custom anomaly detection algorithms, integrated into transaction monitoring pipelines using Python and Azure Functions for scalable deployment.
- Forecasted revenue, stock movement, and financial trends using time series models such as ARIMA, Prophet, and LSTM, enhancing planning accuracy and business forecasting capabilities across departments.
- Performed data wrangling and feature engineering on large-scale financial datasets using Python (Pandas, NumPy), SQL, and PySpark within Azure Databricks, enabling effective model training and evaluation.
- Designed NLP-driven solutions using spaCy and BERT to extract structured insights from annual reports, investor communications, and KYC/AML documentation, improving compliance efficiency and reducing manual processing by 40%.
- Automated ETL workflows using Apache Airflow and dbt integrated with Azure Data Factory, streamlining ingestion and transformation of financial data from multiple internal and third-party sources.
- Deployed ML models to production using MLflow and Docker within Azure ML environments, ensuring reproducibility, performance monitoring, and version control aligned with modern MLOps best practices.
- Collaborated with cross-functional teams including compliance, analysts, and software engineers to convert regulatory and business requirements into interpretable machine learning models using SHAP and LIME for transparency and audit readiness.

Skills

Methodologies: SDLC, Agile, Waterfall.
Language: Python, R, SQL, SAS, DAX, MDX, Java, C++, Scala.
Machine Learning: Regression Analysis, Classification Analysis, Time Series (ARIMA, Prophet, LSTM) Neural Networks (Feedforward, CNN, RNN), NLP, Large Language Models (LLMs), Deep Learning, Feature Engineering, Hyperparameter Tuning, Generative AI (Gen AI), Model Interpretability.
Packages: NumPy, Pandas, Matplotlib, SciPy, ggplot2, Scikit-Learn, PyTorch, TensorFlow, Keras, PySpark.
AI Frameworks & Tools: LangChain, LangGraph, Crew AI, Pydantic AI, MCP, Autogen, Prompt Engineering, Agentic AI, RAG (Retrieval Augmented Generation).
Visualization Tools & IDEs: Tableau, Power BI, JMP, Visual Studio Code, PyCharm, Jupyter Notebook.
Cloud Technologies: AWS (S3, EC2, EMR, ECR, SageMaker, Redshift, CloudWatch), GCP, Azure (Blob Storage, Virtual Machines, AI).
Database: MySQL, SQL Server, PostgreSQL, Oracle, MongoDB.
Statistical Methods & Techniques: Hypothesis Testing, Inferential methods, Causal methods. A/B Testing, Time Series Analysis, Predictive Analytics, Pattern Recognition, Statistical Analysis, Data Modeling, Data Mining, and Exploratory Data Analysis (EDA).
Software & Tools: Jira, Microsoft Excel, Power Automate, Databricks, BigQuery, Snowflake, and SAP.
Operating System: Windows, Mac, Linux.

Education

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA
Bachelor's in Computer Science and Engineering | Vignana Bharathi Institute of Technology, Hyderabad, India